Strict Mathematical Definitions of Function Spaces

This document provides the strict mathematical definitions for a list of classical function and sequence spaces. For each space, we define its elements and its norm (or topology), and then relate it to the three fundamental concepts of **sequences** (convergence of sequences within the space), **continuity** (of the functions that constitute the space), and **Cauchy** (whether the space is complete with respect to its metric, i.e., whether it is a Banach space).

1 Classical L^p Spaces

Definition 1 ($L^p([0,1])$ for $1 \leq p < \infty$)**.** *The space $L^p([0,1])$ consists of all equivalence classes of Lebesgue measurable functions $f : [0,1] \to \mathbb{C}$ for which $|f|^p$ is Lebesgue integrable. Two functions are equivalent if they are equal almost everywhere. The norm is defined as*

$$\|f\|_{L^p} = \left(\int_{[0,1]} |f(x)|^p \, dx \right)^{1/p} < \infty.$$

- ***Sequence:*** *A sequence $\{f_n\}$ converges to f in L^p if $\|f_n - f\|_{L^p} \to 0$. This is known as convergence in p-mean.*
- ***Continuity:*** *The functions themselves are not necessarily continuous; they are equivalence classes of measurable functions.*
- ***Cauchy:*** *$L^p([0,1])$ is complete (a Banach space). Every Cauchy sequence in the L^p norm converges to a limit in the space. For $1 < p < \infty$, it is reflexive.*

Definition 2 ($L^\infty([0,1])$)**.** *The space $L^\infty([0,1])$ consists of all equivalence classes of Lebesgue measurable functions $f : [0,1] \to \mathbb{C}$ that are essentially bounded. The norm is the essential supremum:*

$$\|f\|_{L^\infty} = \operatorname{ess\,sup}_{x \in [0,1]} |f(x)| = \inf\{M \geq 0 : |f(x)| \leq M \text{ for almost every } x\} < \infty.$$

- ***Sequence:*** *$\{f_n\}$ converges to f in L^∞ if $\|f_n - f\|_{L^\infty} \to 0$, meaning $f_n \to f$ uniformly except on a set of measure zero (almost uniform convergence).*
- ***Continuity:*** *Functions are not necessarily continuous; they are essentially bounded.*
- ***Cauchy:*** *$L^\infty([0,1])$ is complete (a Banach space) but is not reflexive.*

Definition 3 ($L^1(\mathbb{R})$)**.** *The space $L^1(\mathbb{R})$ consists of all equivalence classes of Lebesgue measurable functions $f : \mathbb{R} \to \mathbb{C}$ for which $|f|$ is Lebesgue integrable over $\mathbb{R}$. The norm is*

$$\|f\|_{L^1} = \int_{\mathbb{R}} |f(x)| \, dx < \infty.$$

- ***Sequence:*** *$\{f_n\}$ converges to f in L^1 if $\|f_n - f\|_{L^1} \to 0$.*
- ***Continuity:*** *The space is translation invariant: $\|f(x-h) - f(x)\|_{L^1} \to 0$ as $h \to 0$ for $f \in L^1$. The functions themselves are not continuous.*
- ***Cauchy:*** *$L^1(\mathbb{R})$ is complete (a Banach space) but is not reflexive.*

Definition 4 ($L^2(\mathbb{T})$)**.** *Let $\mathbb{T} = \mathbb{R}/2\pi\mathbb{Z}$ be the torus (or circle group). $L^2(\mathbb{T})$ consists of equivalence classes of Lebesgue measurable functions $f : \mathbb{T} \to \mathbb{C}$ such that $|f|^2$ is integrable with respect to the Haar (Lebesgue) measure. The norm is induced by the inner product:*

$$\langle f, g\rangle = \int_{\mathbb{T}} f(x)\overline{g(x)}\,dx, \quad \|f\|_{L^2} = \sqrt{\langle f, f\rangle}.$$

- ***Sequence:*** *Convergence in L^2 is convergence in the mean-square sense.*
- ***Continuity:*** *$L^2(\mathbb{T})$ functions are not necessarily pointwise continuous, but they can be approximated by continuous functions (e.g., Fourier series partial sums converge in L^2).*
- ***Cauchy:*** *$L^2(\mathbb{T})$ is complete, making it a Hilbert space.*

2 Sequence Spaces

Definition 5 ($\ell^p(\mathbb{N})$ for $1 \le p < \infty$)**.** *The space $\ell^p(\mathbb{N})$ consists of all sequences $x = (x_n)_{n\in\mathbb{N}}$ of complex numbers such that*

$$\|x\|_{\ell^p} = \left(\sum_{n=1}^{\infty} |x_n|^p\right)^{1/p} < \infty.$$

- ***Sequence:*** *Convergence of a sequence of sequences $\{x^{(k)}\}$ to x in ℓ^p means $\|x^{(k)} - x\|_{\ell^p} \to 0$.*
- ***Continuity:*** *The "continuity" concept is discrete; we consider boundedness and convergence of the terms x_n.*
- ***Cauchy:*** *ℓ^p is complete (a Banach space) and is reflexive for $1 < p < \infty$.*

Definition 6 ($\ell^\infty(\mathbb{N})$)**.** *The space $\ell^\infty(\mathbb{N})$ consists of all bounded sequences $x = (x_n)_{n\in\mathbb{N}}$ of complex numbers. The norm is the supremum norm:*

$$\|x\|_{\ell^\infty} = \sup_{n\in\mathbb{N}} |x_n| < \infty.$$

- ***Sequence:*** *Convergence in ℓ^∞ is uniform convergence of the sequence entries.*
- ***Continuity:*** *Not applicable; we consider boundedness of the sequence terms.*
- ***Cauchy:*** *ℓ^∞ is complete (a Banach space) but is not reflexive.*

Definition 7 (c_0)**.** *The space c_0 is a closed subspace of ℓ^∞ consisting of all sequences $x = (x_n)$ that converge to 0:*

$$c_0 = \{(x_n) \subset \mathbb{C} : \lim_{n\to\infty} x_n = 0\}, \quad \|x\|_\infty = \sup_n |x_n|.$$

- ***Sequence:*** *$\{x^{(k)}\} \to x$ in c_0 if $\|x^{(k)} - x\|_\infty \to 0$, i.e., uniform convergence of sequences that vanish at infinity.*
- ***Continuity:*** *Not applicable.*
- ***Cauchy:*** *c_0 is complete (a Banach space) and is separable, but its dual is ℓ^1, making it non-reflexive.*

Definition 8 (c)**.** *The space c consists of all convergent sequences $x = (x_n)$ of complex numbers. It is a Banach space with the supremum norm $\|x\|_\infty = \sup_n |x_n|$.*

- ***Sequence:*** *Convergence in c is uniform convergence of sequences that have a (possibly non-zero) limit.*
- ***Continuity:*** *Not applicable.*
- ***Cauchy:*** *c is complete (a Banach space). It is isometrically isomorphic to c_0 via the map $(x_n) \mapsto (x_n - \lim x_n, \lim x_n)$.*

Definition 9 (ℓ^1)**.** *The space $\ell^1(\mathbb{N})$ consists of all absolutely summable sequences. It is a special case of ℓ^p with $p = 1$.*

$$\|x\|_1 = \sum_{n=1}^{\infty} |x_n| < \infty.$$

- ***Sequence:*** *Convergence in ℓ^1 is convergence in the l^1 norm.*
- ***Continuity:*** *Not applicable.*
- ***Cauchy:*** *ℓ^1 is complete (a Banach space) but is non-reflexive. Its dual is ℓ^∞, but the dual of ℓ^∞ is not ℓ^1.*

3 Function Spaces on Domains

Definition 10 ($C[0,1]$)**.** *The space $C[0,1]$ consists of all continuous functions $f : [0,1] \to \mathbb{C}$. The norm is the supremum norm (or uniform norm):*

$$\|f\|_\infty = \sup_{x\in[0,1]} |f(x)|.$$

- ***Sequence:*** *$\{f_n\}$ converges to f in $C[0,1]$ iff $\|f_n - f\|_\infty \to 0$, which is precisely uniform convergence on $[0,1]$.*
- ***Continuity:*** *Every element $f \in C[0,1]$ is, by definition, a uniformly continuous function on a compact set.*
- ***Cauchy:*** *$C[0,1]$ is complete (a Banach space) because the uniform limit of a sequence of continuous functions is continuous.*

Definition 11 ($C_0(\mathbb{R})$)**.** *The space $C_0(\mathbb{R})$ consists of all continuous functions $f : \mathbb{R} \to \mathbb{C}$ that vanish at infinity:*

$$\lim_{|x|\to\infty} f(x) = 0.$$

It is equipped with the supremum norm $\|f\|_\infty = \sup_{x\in\mathbb{R}} |f(x)|$.

- ***Sequence:*** *$\{f_n\}$ converges to f in $C_0(\mathbb{R})$ iff $\sup_{x\in\mathbb{R}} |f_n(x) - f(x)| \to 0$ (uniform convergence on the whole real line).*
- ***Continuity:*** *Every element is continuous and vanishes at infinity.*
- ***Cauchy:*** *$C_0(\mathbb{R})$ is complete (a Banach space).*

Definition 12 ($C_b(\mathbb{R})$)**.** *The space $C_b(\mathbb{R})$ consists of all bounded continuous functions $f : \mathbb{R} \to \mathbb{C}$. The norm is the supremum norm $\|f\|_\infty = \sup_{x\in\mathbb{R}} |f(x)|$.*

- ***Sequence:*** *Convergence in $C_b(\mathbb{R})$ is uniform convergence on $\mathbb{R}$.*
- ***Continuity:*** *Every element is continuous and bounded.*
- ***Cauchy:*** *$C_b(\mathbb{R})$ is complete (a Banach space).*

Definition 13 ($C^k[0,1]$)**.** *The space $C^k[0,1]$ for $k \in \mathbb{N}$ consists of all k-times continuously differentiable functions on $[0,1]$. The norm is often defined as the sum of the uniform norms of the function and its derivatives:*

$$\|f\|_{C^k} = \sum_{j=0}^{k} \|f^{(j)}\|_\infty.$$

- ***Sequence:*** *$\{f_n\}$ converges to f in C^k iff $f_n^{(j)} \to f^{(j)}$ uniformly on $[0,1]$ for all $j = 0, 1, \ldots, k$.*
- ***Continuity:*** *Every element and its derivatives up to order k are continuous.*
- ***Cauchy:*** *$C^k[0,1]$ is complete (a Banach space).*

4 Sobolev Spaces

Let $\Omega \subset \mathbb{R}^n$ be a domain.

Definition 14 ($W^{1,p}(\Omega)$ for $1 \leq p \leq \infty$)**.** *The space $W^{1,p}(\Omega)$ consists of all functions $f \in L^p(\Omega)$ whose weak (distributional) partial derivatives $D^\alpha f$ (for $|\alpha| \leq 1$) exist and belong to $L^p(\Omega)$. The norm is*

$$\|f\|_{W^{1,p}} = \|f\|_{L^p} + \sum_{|\alpha| \leq 1} \|D^\alpha f\|_{L^p}.$$

- ***Sequence:*** *Convergence in $W^{1,p}$ means convergence of the functions and their weak derivatives in the L^p norm.*
- ***Continuity:*** *For low p and high dimension, elements are not necessarily continuous in the classical sense (Sobolev embedding theorems give conditions for continuity).*
- ***Cauchy:*** *$W^{1,p}(\Omega)$ is complete (a Banach space) by definition; it is the completion of $C^\infty(\Omega)$ in this norm.*

Definition 15 ($W_0^{1,p}(\Omega)$)**.** *The space $W_0^{1,p}(\Omega)$ is the closure of $C_c^\infty(\Omega)$ (infinitely differentiable functions with compact support in Ω) in the $W^{1,p}(\Omega)$ norm. It represents functions that "vanish" on the boundary $\partial\Omega$ in the trace sense.*

- ***Sequence:*** *Every element of $W_0^{1,p}$ is the limit in the $W^{1,p}$ norm of a sequence of smooth, compactly supported functions.*
- ***Continuity:*** *Same continuity properties as $W^{1,p}$, but with zero boundary values.*
- ***Cauchy:*** *$W_0^{1,p}(\Omega)$ is complete (a Banach space) as it is a closed subspace of $W^{1,p}(\Omega)$.*

Definition 16 ($H^1(\Omega)$ and $H_0^1(\Omega)$)**.** *We define $H^1(\Omega) = W^{1,2}(\Omega)$ and $H_0^1(\Omega) = W_0^{1,2}(\Omega)$. These are Hilbert spaces with the inner product*

$$\langle u, v \rangle_{H^1} = \int_\Omega uv \, dx + \int_\Omega \nabla u \cdot \nabla v \, dx.$$

- ***Sequence:*** *Convergence in H^1 implies L^2 convergence of the functions and their gradients.*
- ***Continuity:*** *$H^1(\Omega)$ is the "energy space" for elliptic PDEs like $-\Delta u = f$.*
- ***Cauchy:*** *Both H^1 and H_0^1 are complete (Hilbert spaces). H_0^1 embeds compactly into $L^2(\Omega)$ for bounded Ω (Rellich-Kondrachov theorem).*

5 Operator and Geometric Spaces

Definition 17 ($B(H)$)**.** *Let H be a Hilbert space. $B(H)$ is the space of all bounded linear operators $T : H \to H$. The norm is the operator norm:*

$$\|T\|_{B(H)} = \sup_{\|x\|_H = 1} \|Tx\|_H.$$

- ***Sequence:*** *$\{T_n\}$ converges to T in $B(H)$ if $\|T_n - T\|_{B(H)} \to 0$, known as uniform operator convergence.*
- ***Continuity:*** *Not applicable; we consider boundedness.*
- ***Cauchy:*** *$B(H)$ is complete (a Banach space). It is not separable in general.*

Definition 18 ($K(H)$)**.** *$K(H) \subset B(H)$ is the space of compact operators on H. An operator T is compact if the image of the unit ball under T is pre-compact in H (i.e., has compact closure).*

- ***Sequence:*** *$K(H)$ is the closure of the space of finite-rank operators in the operator norm of $B(H)$. A sequence of compact operators converging uniformly to T implies T is compact.*

- ***Continuity:*** *Not applicable.*
- ***Cauchy:*** *$K(H)$ is a closed ideal in $B(H)$, and thus complete.*

Definition 19 ($C_1(H)$ (Trace Class))**.** *$C_1(H)$ is the space of trace-class operators on H. An operator $T \in K(H)$ is trace-class if for its singular values s_n (eigenvalues of $|T| = \sqrt{T^*T}$), we have $\sum s_n < \infty$. The norm is the trace norm $\|T\|_{C_1} = \mathrm{tr}(|T|)$.*

- ***Sequence:*** *Convergence in trace norm implies compactness and is stronger than operator norm convergence.*
- ***Continuity:*** *Not applicable.*
- ***Cauchy:*** *$C_1(H)$ is complete (a Banach space).*

Definition 20 (Lip([0, 1]))**.** *The space Lip([0, 1]) consists of Lipschitz continuous functions on $[0, 1]$, i.e., functions $f : [0, 1] \to \mathbb{C}$ for which there exists a constant L such that $|f(x) - f(y)| \leq L|x - y|$ for all $x, y \in [0, 1]$. It is a Banach space with the norm*

$$\|f\|_{Lip} = \|f\|_\infty + \sup_{x \neq y} \frac{|f(x) - f(y)|}{|x - y|}.$$

- ***Sequence:*** *$\{f_n\}$ converges to f in Lip iff $f_n \to f$ uniformly and their Lipschitz constants are uniformly bounded and converge.*
- ***Continuity:*** *Every element is Lipschitz continuous, hence absolutely continuous and differentiable almost everywhere.*
- ***Cauchy:*** *Lip([0, 1]) is complete (a Banach space).*

6 Mixed and Special Spaces

Definition 21 ($C^{0,\alpha}([0, 1])$, Hölder Spaces)**.** *For $0 < \alpha \leq 1$, $C^{0,\alpha}([0, 1])$ consists of functions f that are Hölder continuous of exponent α:*

$$\|f\|_{C^{0,\alpha}} = \|f\|_\infty + \sup_{x \neq y} \frac{|f(x) - f(y)|}{|x - y|^\alpha} < \infty.$$

For $\alpha = 1$, this is the Lipschitz space.

- ***Sequence:*** *Convergence in $C^{0,\alpha}$ implies uniform convergence and convergence of the Hölder seminorm.*
- ***Continuity:*** *Functions are uniformly continuous.*
- ***Cauchy:*** *$C^{0,\alpha}([0, 1])$ is complete (a Banach space).*

Definition 22 ($BV([0, 1])$)**.** *$BV([0, 1])$ is the space of functions of bounded variation on $[0, 1]$. A function f is in BV if its total variation $V_0^1(f)$ is finite:*

$$V_0^1(f) = \sup_P \sum_{i=1}^n |f(x_i) - f(x_{i-1})| < \infty,$$

where the supremum is taken over all partitions $P = \{0 = x_0 < x_1 < \cdots < x_n = 1\}$. The norm is $\|f\|_{BV} = \|f\|_{L^1} + V_0^1(f)$.

- ***Sequence:*** *Convergence in BV is strong.*
- ***Continuity:*** *BV functions can have jump discontinuities; they are not necessarily continuous.*
- ***Cauchy:*** *$BV([0, 1])$ is complete (a Banach space).*

Definition 23 ($L^p(\mu)$ on a probability space)**.** *Let $(\Omega, \mathcal{F}, \mu)$ be a probability space ($\mu(\Omega) = 1$). $L^p(\mu)$ for $1 \leq p < \infty$ is the space of equivalence classes of measurable functions $f : \Omega \to \mathbb{C}$ such that $\int_\Omega |f|^p \, d\mu < \infty$, with norm $\|f\|_{L^p} = (\int_\Omega |f|^p d\mu)^{1/p}$. $L^\infty(\mu)$ is the space of essentially bounded functions.*

- ***Sequence:*** *Convergence in $L^p(\mu)$ is convergence in p-mean with respect to the measure μ.*
- ***Continuity:*** *The concept of pointwise continuity is replaced by measurability.*
- ***Cauchy:*** *$L^p(\mu)$ is complete (a Banach space) for all $1 \leq p \leq \infty$.*

Definition 24 ($C(\mathbb{T})$)**.** *$\mathbb{T} = \{z \in \mathbb{C} : |z| = 1\}$ is the circle group. $C(\mathbb{T})$ consists of all continuous functions $f : \mathbb{T} \to \mathbb{C}$. It is isometrically isomorphic to the space of continuous 2π-periodic functions on $\mathbb{R}$. The norm is the supremum norm $\|f\|_\infty = \sup_{z \in \mathbb{T}} |f(z)|$.*

- ***Sequence:*** *Convergence in $C(\mathbb{T})$ is uniform convergence on the circle.*
- ***Continuity:*** *Every element is continuous on the compact set $\mathbb{T}$, hence uniformly continuous.*
- ***Cauchy:*** *$C(\mathbb{T})$ is complete (a Banach space).*

7 Hardy and Analytic Spaces

Let $\mathbb{D} = \{z \in \mathbb{C} : |z| < 1\}$ be the open unit disk and $\mathbb{T} = \partial\mathbb{D}$ the unit circle.

Definition 25 ($H^p(\mathbb{D})$ – Hardy Spaces on the Unit Disk)**.** *For $0 < p < \infty$, the Hardy space $H^p(\mathbb{D})$ consists of all analytic functions $f : \mathbb{D} \to \mathbb{C}$ such that*

$$\|f\|_{H^p} = \sup_{0 \leq r < 1} \left(\frac{1}{2\pi} \int_0^{2\pi} |f(re^{i\theta})|^p \, d\theta \right)^{1/p} < \infty.$$

For $p = \infty$, $H^\infty(\mathbb{D})$ consists of bounded analytic functions on $\mathbb{D}$ with $\|f\|_{H^\infty} = \sup_{z \in \mathbb{D}} |f(z)|$.

- ***Sequence:*** *A sequence $\{f_n\}$ converges to f in H^p if $\|f_n - f\|_{H^p} \to 0$. This implies uniform convergence on compact subsets of $\mathbb{D}$.*
- ***Continuity:*** *Every element is analytic, hence infinitely differentiable in the complex sense (holomorphic).*
- ***Cauchy:*** *$H^p(\mathbb{D})$ is complete (a Banach space for $p \geq 1$, a Fréchet space for $p < 1$). For $1 < p < \infty$, it is reflexive.*

Definition 26 ($H^1(\mathbb{T})$ – Boundary Values of Hardy Functions)**.** *$H^1(\mathbb{T})$ can be identified with the subspace of $L^1(\mathbb{T})$ consisting of functions whose Fourier coefficients $\hat{f}(n)$ vanish for all $n < 0$. These are precisely the boundary values of functions in $H^1(\mathbb{D})$, which exist as L^1 functions on $\mathbb{T}$ for radial limits almost everywhere.*

$$H^1(\mathbb{T}) = \{f \in L^1(\mathbb{T}) : \hat{f}(n) = 0 \text{ for all } n < 0\}.$$

- ***Sequence:*** *$\{f_n\} \to f$ in $H^1(\mathbb{T})$ iff $\|f_n - f\|_{L^1(\mathbb{T})} \to 0$ and the analyticity condition (vanishing negative Fourier coefficients) holds for each f_n.*
- ***Continuity:*** *Functions are not necessarily continuous; they are L^1 functions with a specific Fourier series structure.*
- ***Cauchy:*** *$H^1(\mathbb{T})$ is complete (a Banach space) as a closed subspace of $L^1(\mathbb{T})$. Its dual is $BMO(\mathbb{T})$ (bounded mean oscillation).*

Definition 27 ($H^\infty(\mathbb{D})$ – Bounded Analytic Functions)**.** *$H^\infty(\mathbb{D})$ consists of all bounded analytic functions $f : \mathbb{D} \to \mathbb{C}$ with norm*

$$\|f\|_{H^\infty} = \sup_{z \in \mathbb{D}} |f(z)| < \infty.$$

- ***Sequence:*** *Convergence in H^∞ is uniform convergence on the entire disk $\mathbb{D}$.*
- ***Continuity:*** *Functions are analytic (hence continuous) and uniformly bounded.*
- ***Cauchy:*** *$H^\infty(\mathbb{D})$ is complete (a Banach algebra) but is non-separable and non-reflexive.*

Definition 28 ($A^2(\mathbb{D})$ – Bergman Space)**.** *The Bergman space $A^2(\mathbb{D})$ consists of all analytic functions $f : \mathbb{D} \to \mathbb{C}$ that are square-integrable with respect to the Lebesgue area measure $dA = \frac{1}{\pi} dx\, dy = \frac{1}{\pi} r\, dr\, d\theta$:*

$$\|f\|_{A^2} = \left(\int_{\mathbb{D}} |f(z)|^2 \, dA \right)^{1/2} < \infty.$$

It is a Hilbert space with inner product $\langle f, g \rangle = \int_{\mathbb{D}} f(z)\overline{g(z)}\, dA$.

- ***Sequence:*** *$\{f_n\} \to f$ in A^2 iff $\int_{\mathbb{D}} |f_n - f|^2 dA \to 0$, which implies uniform convergence on compact subsets.*
- ***Continuity:*** *Elements are analytic, hence C^∞.*
- ***Cauchy:*** *$A^2(\mathbb{D})$ is complete (a Hilbert space) and is separable. The evaluation functionals $f \mapsto f(z)$ are continuous (reproducing kernel Hilbert space).*

8 Besov and Triebel-Lizorkin Spaces

These spaces are typically defined on $\mathbb{R}^n$ using the Fourier transform and dyadic partitions of unity. Let $\{\varphi_j\}_{j=0}^\infty$ be a smooth dyadic resolution of unity.

Definition 29 ($B^s_{p,q}(\mathbb{R}^n)$ – Besov Spaces)**.** *For $s \in \mathbb{R}$, $0 < p, q \leq \infty$, the Besov space $B^s_{p,q}(\mathbb{R}^n)$ consists of all tempered distributions f such that*

$$\|f\|_{B^s_{p,q}} = \left(\sum_{j=0}^\infty 2^{jsq} \|\mathcal{F}^{-1}(\varphi_j \mathcal{F} f)\|^q_{L^p} \right)^{1/q} < \infty,$$

with the usual modification for $q = \infty$. Here $\mathcal{F}$ denotes the Fourier transform.

- ***Sequence:*** *Convergence in $B^s_{p,q}$ means convergence of the sequence of dyadic frequency blocks in the appropriate $\ell^q(L^p)$ norm.*
- ***Continuity:*** *For sufficiently large $s > n/p$, elements are continuous functions (Sobolev embedding). In general, they are distributions.*
- ***Cauchy:*** *$B^s_{p,q}(\mathbb{R}^n)$ is complete (a quasi-Banach space for $p, q < 1$, a Banach space for $p, q \geq 1$). They are the natural spaces for wavelet analysis.*

Definition 30 ($F^s_{p,q}(\mathbb{R}^n)$ – Triebel-Lizorkin Spaces)**.** *For $s \in \mathbb{R}$, $0 < p < \infty$, $0 < q \leq \infty$, the Triebel-Lizorkin space $F^s_{p,q}(\mathbb{R}^n)$ consists of all tempered distributions f such that*

$$\|f\|_{F^s_{p,q}} = \left\| \left(\sum_{j=0}^\infty 2^{jsq} |\mathcal{F}^{-1}(\varphi_j \mathcal{F} f)(\cdot)|^q \right)^{1/q} \right\|_{L^p} < \infty,$$

with the usual modification for $q = \infty$.

- ***Sequence:*** *Convergence in $F^s_{p,q}$ means convergence of the L^p norm of the ℓ^q-sum of the frequency blocks.*
- ***Continuity:*** *Similar to Besov spaces; they unify many classical spaces (e.g., $F^s_{p,2} = H^s_p$, the Bessel-potential Sobolev spaces).*

- ***Cauchy:*** *$F^s_{p,q}(\mathbb{R}^n)$ is complete (a quasi-Banach or Banach space).*

Definition 31 ($B^s_{2,2}(\Omega)$)**.** *For $s \in \mathbb{R}$ and a suitable domain $\Omega \subset \mathbb{R}^n$, the space $B^s_{2,2}(\Omega)$ is equivalent to the Sobolev space $H^s(\Omega) = W^{s,2}(\Omega)$ (the L^2-Sobolev space of order s). On $\mathbb{R}^n$, we have the norm equivalence:*

$$\|f\|_{B^s_{2,2}} \approx \|f\|_{H^s} = \left(\int_{\mathbb{R}^n} (1 + |\xi|^2)^s |\hat{f}(\xi)|^2 \, d\xi \right)^{1/2}.$$

- ***Sequence:*** *Convergence in $B^s_{2,2}$ is equivalent to convergence in the L^2-Sobolev norm.*
- ***Continuity:*** *By Sobolev embedding, if $s > n/2$, functions are continuous.*
- ***Cauchy:*** *$B^s_{2,2}$ is a Hilbert space (complete).*

Definition 32 ($W^{s,p}(\Omega)$ – Slobodeckij Spaces, $0 < s < 1$)**.** *For $0 < s < 1$ and $1 \leq p < \infty$, the Slobodeckij (fractional Sobolev) space $W^{s,p}(\Omega)$ consists of functions $f \in L^p(\Omega)$ such that the Gagliardo (Slobodeckij) semi-norm is finite:*

$$[f]_{W^{s,p}} = \left(\int_\Omega \int_\Omega \frac{|f(x) - f(y)|^p}{|x - y|^{n+sp}} \, dx \, dy \right)^{1/p} < \infty.$$

The norm is $\|f\|_{W^{s,p}} = (\|f\|^p_{L^p} + [f]^p_{W^{s,p}})^{1/p}$.

- ***Sequence:*** *$\{f_n\} \to f$ in $W^{s,p}$ iff $\|f_n - f\|_{L^p} \to 0$ and $[f_n - f]_{W^{s,p}} \to 0$.*
- ***Continuity:*** *For $sp > n$, functions are continuous (Morrey's inequality). For $sp \leq n$, they are generally not continuous.*
- ***Cauchy:*** *$W^{s,p}(\Omega)$ is complete (a Banach space).*

9 Operator Spaces

Definition 33 ($B(X)$)**.** *Let X be a Banach space. $B(X)$ is the space of all bounded linear operators $T : X \to X$. The norm is the operator norm:*

$$\|T\|_{B(X)} = \sup_{\|x\|_X = 1} \|Tx\|_X.$$

- ***Sequence:*** *$\{T_n\} \to T$ in $B(X)$ iff $\|T_n - T\|_{B(X)} \to 0$ (uniform operator topology).*
- ***Continuity:*** *Not applicable; we consider boundedness.*
- ***Cauchy:*** *$B(X)$ is complete (a Banach algebra). It is unital if X is non-trivial.*

Definition 34 ($B(H, K)$)**.** *Let H and K be Hilbert spaces. $B(H, K)$ is the space of all bounded linear operators $T : H \to K$ with the operator norm*

$$\|T\| = \sup_{\|x\|_H = 1} \|Tx\|_K.$$

When $H = K$, we denote $B(H) = B(H, H)$.

- ***Sequence:*** *Convergence in $B(H, K)$ is uniform convergence on the unit ball of H.*
- ***Continuity:*** *Not applicable.*
- ***Cauchy:*** *$B(H, K)$ is complete (a Banach space).*

Definition 35 (Schatten Class $S^p(H)$)**.** *For $1 \le p < \infty$, the Schatten class $S^p(H)$ consists of all compact operators $T \in B(H)$ such that the sequence of singular values $s_n(T) = \lambda_n(|T|)$ (where $|T| = \sqrt{T^*T}$) is in ℓ^p. The norm is*

$$\|T\|_{S^p} = \left(\sum_{n=1}^{\infty} s_n(T)^p \right)^{1/p} = (\mathrm{tr}(|T|^p))^{1/p}.$$

For $p = \infty$, $S^\infty(H) = K(H)$ with the operator norm. For $p = 2$, $S^2(H)$ is the Hilbert space of Hilbert-Schmidt operators.

- ***Sequence:*** *$\{T_n\} \to T$ in S^p iff $\|T_n - T\|_{S^p} \to 0$.*
- ***Continuity:*** *These are "non-commutative L^p spaces" for the von Neumann algebra $B(H)$.*
- ***Cauchy:*** *$S^p(H)$ is complete (a Banach space for $p \ge 1$, a Banach algebra only for $p = \infty$). For $1 < p < \infty$, it is reflexive.*

Definition 36 (Nuclear Operators $\mathcal{N}(X, Y)$)**.** *Let X and Y be Banach spaces. An operator $T \in B(X, Y)$ is nuclear if there exist sequences $\{x'_n\} \subset X'$ and $\{y_n\} \subset Y$ such that $\sum_{n=1}^{\infty} \|x'_n\|\|y_n\| < \infty$ and*

$$Tx = \sum_{n=1}^{\infty} x'_n(x) y_n \quad \text{for all } x \in X.$$

The nuclear norm is $\|T\|_{\mathcal{N}} = \inf \sum \|x'_n\|\|y_n\|$, where the infimum is taken over all such representations.

- ***Sequence:*** *Convergence in nuclear norm implies convergence in operator norm.*
- ***Continuity:*** *Nuclear operators are compact.*
- ***Cauchy:*** *$\mathcal{N}(X, Y)$ is complete (a Banach space). On Hilbert spaces, $\mathcal{N}(H) = S^1(H)$ (trace class).*

10 Classical Geometric Spaces

Definition 37 (James Space J)**.** *The James space J is the space of all sequences $x = (x_n)_{n=1}^{\infty}$ of real numbers such that $\lim_{n\to\infty} x_n = 0$ and*

$$\|x\|_J = \sup \left(\sum_{k=1}^{n-1} |x_{p_{k+1}} - x_{p_k}|^2 \right)^{1/2} < \infty,$$

where the supremum is taken over all finite increasing sequences $p_1 < p_2 < \cdots < p_n$ of indices.

- ***Sequence:*** *A sequence $\{x^{(k)}\}$ converges to x in J if $\|x^{(k)} - x\|_J \to 0$.*
- ***Continuity:*** *Not applicable; we consider sequences.*
- ***Cauchy:*** *J is complete (a Banach space). It is non-reflexive but isometrically isomorphic to its double dual J^{**} (i.e., it is "almost reflexive").*

Definition 38 (Tsirelson Space T)**.** *The Tsirelson space T is the completion of c_{00} (finitely supported sequences) under the implicit norm:*

$$\|x\|_T = \max \left\{ \|x\|_{\ell^\infty}, \frac{1}{2} \sup_{n \le E_1 < \cdots < E_n} \sum_{i=1}^{n} \|E_i x\|_T \right\},$$

where E_i are successive finite subsets of $\mathbb{N}$ and $E_i x$ denotes the restriction of x to E_i. It is the first example of a Banach space containing no copy of c_0 or ℓ^p for any $1 \le p < \infty$.

- ***Sequence:*** *Defined by the completion process.*

- ***Continuity:*** *Not applicable.*
- ***Cauchy:*** *T is complete by construction. It does not have any cotype (i.e., it is not of finite cotype).*

Definition 39 (Schreier Space S)**.** *The Schreier space S is the completion of c_{00} under the norm*

$$\|x\|_S = \sup\left\{\sum_{k=1}^{n} |x_{n_k}| : n \le n_1 < \cdots < n_n\right\},$$

where the supremum is taken over all finite sets $\{n_1, \ldots, n_n\}$ with $n \le n_1$.

- ***Sequence:*** *Convergence in S is defined by this norm.*
- ***Continuity:*** *Not applicable.*
- ***Cauchy:*** *S is complete (a Banach space). It is distortable, meaning it can be renormed to be arbitrarily "far" from being isometric to a Hilbert space.*

Definition 40 (Dvoretzky-Rogers Space)**.** *Typically refers to a construction (e.g., $\ell^2 \cap \ell^1$) that demonstrates the Dvoretzky-Rogers theorem: every infinite-dimensional Banach space has an unconditional basis that is not absolutely summing. A concrete example is the space of sequences with norm*

$$\|x\| = \max\left\{\|x\|_2, \frac{1}{\sqrt{2}}\|x\|_1\right\}.$$

- ***Sequence:*** *Convergence is defined by this mixed norm.*
- ***Continuity:*** *Not applicable.*
- ***Cauchy:*** *This construction yields a complete space (a Banach space) with an unconditional basis that is not absolutely summing.*

11 Weighted and Orlicz Spaces

Definition 41 (Weighted $L^p_w(\mathbb{R}^n)$)**.** *Let $w : \mathbb{R}^n \to [0, \infty]$ be a measurable, almost everywhere positive weight function. The weighted Lebesgue space $L^p_w(\mathbb{R}^n)$ for $1 \le p < \infty$ consists of measurable functions f such that*

$$\|f\|_{L^p_w} = \left(\int_{\mathbb{R}^n} |f(x)|^p w(x)\, dx\right)^{1/p} < \infty.$$

For $p = \infty$, $L^\infty_w(\mathbb{R}^n)$ is the space of measurable functions with $\|f\|_{L^\infty_w} = \operatorname{ess\,sup}_x w(x)^{1/p}|f(x)| < \infty$.

- ***Sequence:*** *$\{f_n\} \to f$ in L^p_w iff $\int |f_n - f|^p w\, dx \to 0$.*
- ***Continuity:*** *Not required.*
- ***Cauchy:*** *$L^p_w(\mathbb{R}^n)$ is complete (a Banach space) for any $p \ge 1$.*

Definition 42 (Orlicz Space $L^\Phi(\Omega)$)**.** *Let $\Phi : [0, \infty) \to [0, \infty)$ be a Young function (convex, $\Phi(0) = 0$, $\lim_{t\to\infty} \Phi(t)/t = \infty$). The Orlicz space $L^\Phi(\Omega)$ consists of measurable functions $f : \Omega \to \mathbb{C}$ such that the Luxemburg norm is finite:*

$$\|f\|_\Phi = \inf\left\{\lambda > 0 : \int_\Omega \Phi\left(\frac{|f(x)|}{\lambda}\right) dx \le 1\right\} < \infty.$$

- ***Sequence:*** *Convergence in L^Φ is convergence in the Luxemburg norm.*
- ***Continuity:*** *Not required.*

- ***Cauchy:*** *$L^\Phi(\Omega)$ is complete (a Banach space) when Φ satisfies the Δ_2 condition (for infinite measure) or always for finite measure.*

Definition 43 (Zygmund Space $C^*(\mathbb{R})$)**.** *The Zygmund space $C^*(\mathbb{R})$ (sometimes denoted $\Lambda_*(\mathbb{R})$) consists of continuous functions f on $\mathbb{R}$ such that*

$$\|f\|_{C^*} = \|f\|_\infty + \sup_{x,h\neq 0} \frac{|f(x+h)+f(x-h)-2f(x)|}{|h|} < \infty.$$

It is a "logarithmic Hölder" space, closely related to C^1 and Lipschitz spaces.

- ***Sequence:*** *$\{f_n\} \to f$ in C^* iff $\|f_n - f\|_\infty \to 0$ and the second difference quotients converge uniformly.*
- ***Continuity:*** *Every element is continuous. The condition controls the modulus of continuity.*
- ***Cauchy:*** *$C^*(\mathbb{R})$ is complete (a Banach space).*

Definition 44 (Lorentz Space $\Lambda^p_w([0,1])$)**.** *Let w be a positive decreasing weight function on $[0,1]$ (e.g., $w(t) = t^{p/q-1}$). The Lorentz space $\Lambda^p_w([0,1])$ consists of measurable functions f such that*

$$\|f\|_{\Lambda^p_w} = \left(\int_0^1 f^*(t)^p w(t)\,dt\right)^{1/p} < \infty,$$

where f^* is the decreasing rearrangement of f. The more common notation is $L^{p,q}$.

- ***Sequence:*** *Convergence in Lorentz spaces is defined by this norm.*
- ***Continuity:*** *Not required.*
- ***Cauchy:*** *$\Lambda^p_w([0,1])$ is complete (a Banach space) for $1 \le p < \infty$ under certain conditions on w (e.g., w being a power function).*

12 Spaces on Manifolds and Groups

Definition 45 ($C_c^\infty(M)$)**.** *Let M be a smooth manifold. $C_c^\infty(M)$ is the space of smooth (infinitely differentiable) functions $f : M \to \mathbb{C}$ with compact support. It is a Fréchet space when endowed with the topology of uniform convergence of all derivatives on compact sets, but it is not normable.*

- ***Sequence:*** *$\{f_n\} \to f$ in $C_c^\infty(M)$ iff there exists a compact $K \subset M$ containing all supports, and all derivatives converge uniformly on K.*
- ***Continuity:*** *Elements are smooth (C^∞).*
- ***Cauchy:*** *$C_c^\infty(M)$ is sequentially complete (a Montel space) but not a Banach space.*

Definition 46 ($W^{k,p}(M)$ – Sobolev Spaces on Riemannian Manifolds)**.** *Let (M,g) be a compact Riemannian manifold. $W^{k,p}(M)$ is the completion of $C^\infty(M)$ with respect to the norm*

$$\|f\|_{W^{k,p}} = \left(\sum_{j=0}^{k} \int_M |\nabla^j f|^p \, dV_g\right)^{1/p},$$

where ∇^j is the j-th covariant derivative and dV_g is the Riemannian volume form.

- ***Sequence:*** *Convergence in $W^{k,p}$ means convergence of the function and its covariant derivatives up to order k in L^p.*
- ***Continuity:*** *By Sobolev embedding theorems on manifolds, for $kp > \dim M$, elements are continuous.*
- ***Cauchy:*** *$W^{k,p}(M)$ is complete (a Banach space, Hilbert for $p = 2$).*

Definition 47 ($L^p(G)$ for Non-Abelian Groups)**.** *Let G be a locally compact (possibly non-abelian) group with left Haar measure μ. $L^p(G)$ for $1 \leq p < \infty$ is the space of equivalence classes of measurable functions $f : G \to \mathbb{C}$ such that*

$$\|f\|_{L^p} = \left(\int_G |f(g)|^p \, d\mu(g) \right)^{1/p} < \infty.$$

$L^\infty(G)$ is defined analogously. The Peter-Weyl theorem states that $L^2(G)$ decomposes as a Hilbert direct sum of irreducible representations.

- ***Sequence:*** *$\{f_n\} \to f$ in L^p iff $\|f_n - f\|_{L^p} \to 0$.*
- ***Continuity:*** *Not required.*
- ***Cauchy:*** *$L^p(G)$ is complete (a Banach space) for all $1 \leq p \leq \infty$.*

Definition 48 ($L^2(\mathbb{R}^n, d\mu)$ with Gaussian Measure)**.** *Consider $\mathbb{R}^n$ with the Gaussian probability measure $d\mu(x) = (2\pi)^{-n/2} e^{-|x|^2/2} dx$. $L^2(\mathbb{R}^n, d\mu)$ consists of measurable functions f such that*

$$\|f\|_{L^2(d\mu)} = \left(\int_{\mathbb{R}^n} |f(x)|^2 \, d\mu(x) \right)^{1/2} < \infty.$$

It is a Hilbert space with inner product $\langle f, g \rangle = \int_{\mathbb{R}^n} f(x)\overline{g(x)} \, d\mu(x)$.

- ***Sequence:*** *Convergence in this space is mean-square convergence with respect to Gaussian measure.*
- ***Continuity:*** *Not required.*
- ***Cauchy:*** *$L^2(\mathbb{R}^n, d\mu)$ is complete (a Hilbert space). It is the setting for the Ornstein-Uhlenbeck operator and Malliavin calculus.*

13 Miscellaneous Key Spaces

Definition 49 ($C_0(G)$)**.** *Let G be a locally compact Hausdorff group. $C_0(G)$ is the space of continuous functions $f : G \to \mathbb{C}$ that vanish at infinity:*

$$\forall \epsilon > 0, \exists K \subset G \text{ compact such that } \sup_{x \notin K} |f(x)| < \epsilon.$$

The norm is the supremum norm $\|f\|_\infty = \sup_{x \in G} |f(x)|$.

- ***Sequence:*** *$\{f_n\} \to f$ in $C_0(G)$ iff $\sup_{x \in G} |f_n(x) - f(x)| \to 0$.*
- ***Continuity:*** *Elements are continuous and vanish at infinity.*
- ***Cauchy:*** *$C_0(G)$ is complete (a Banach space). It is the closure of $C_c(G)$ (compactly supported continuous functions) in the sup norm.*

Definition 50 (Marcinkiewicz Space $M^p([0,1])$ (Weak L^p))**.** *For $1 \leq p < \infty$, the Marcinkiewicz space (weak L^p) consists of measurable functions f such that*

$$\|f\|_{M^p} = \sup_{t > 0} t \cdot \lambda_f(t)^{1/p} < \infty,$$

where $\lambda_f(t) = |\{x \in [0,1] : |f(x)| > t\}|$ is the distribution function. The quasinorm satisfies $\|f\|_{M^p} \leq \|f\|_{L^p}$.

- ***Sequence:*** *Convergence is defined by this quasinorm, often with the topology of convergence in measure.*
- ***Continuity:*** *Not required.*

- ***Cauchy:*** *$M^p([0,1])$ is complete (a quasi-Banach space). It is larger than L^p and is the natural domain for many interpolation theorems.*

Definition 51 (BMO($\mathbb{R}^n$) – Bounded Mean Oscillation)**.** *BMO($\mathbb{R}^n$) consists of all locally integrable functions f such that*

$$\|f\|_{BMO} = \sup_{Q \subset \mathbb{R}^n} \frac{1}{|Q|} \int_Q |f(x) - f_Q| \, dx < \infty,$$

where the supremum is over all cubes $Q \subset \mathbb{R}^n$, $|Q|$ is the Lebesgue measure, and $f_Q = \frac{1}{|Q|} \int_Q f(x) \, dx$ is the average of f over Q. Functions differing by a constant are identified.

- ***Sequence:*** *A sequence $\{f_n\}$ converges to f in BMO if $\|f_n - f\|_{BMO} \to 0$.*
- ***Continuity:*** *BMO functions can be discontinuous (e.g., $\log|x| \in$ BMO). They are defined modulo constants.*
- ***Cauchy:*** *BMO($\mathbb{R}^n$) is complete (a Banach space). It is the dual of the Hardy space $H^1(\mathbb{R}^n)$.*

Definition 52 (VMO($\mathbb{R}^n$) – Vanishing Mean Oscillation)**.** *VMO($\mathbb{R}^n$) is the subspace of BMO($\mathbb{R}^n$) consisting of functions with vanishing mean oscillation:*

$$\lim_{a \to 0^+} \sup_{Q: diam(Q) < a} \frac{1}{|Q|} \int_Q |f(x) - f_Q| \, dx = 0,$$

and also the condition as cubes tend to infinity (for unbounded domains). It is the closure of $C_c^\infty(\mathbb{R}^n)$ (or uniformly continuous functions) in the BMO norm.

- ***Sequence:*** *$\{f_n\} \to f$ in VMO iff $\|f_n - f\|_{BMO} \to 0$ and each f_n (and the limit) satisfies the vanishing condition.*
- ***Continuity:*** *VMO functions can be approximated by smooth functions; they have mean oscillation that decays with the scale.*
- ***Cauchy:*** *VMO($\mathbb{R}^n$) is complete (a Banach space) as a closed subspace of BMO.*